UNIVERSITY OF DERBY

College of Science and Engineering

Practice Paper B Solutions

OPERATIONAL RESEARCH – 5CM522

1. Scheduling

(a) Applying Johnson's Rule gives the sequence

$$J_4,\ J_2,\ J_5,\ J_1,\ J_3$$

Main steps:

Step	Smallest time remaining	Placement rule	Partial sequence
1	2 on J_4 for M_1	place first	$J_4, _, _, _, _$
2	3 on J_2 for M_1	next from front	$J_4, J_2, _, _, _$
3	4 on J_3 for M_2	place last	$J_4, J_2, _, _, J_3$
4	5 on J_1 for M_2	next from back	$J_4, J_2, _, J_1, J_3$
5	5 on J_5 for M_1	fill remaining front slot	J_4, J_2, J_5, J_1, J_3

(b) Gantt timings:

Job	Machine 1	Machine 2
J_4	0–2	2–11
J_2	2–5	11–19
J_5	5–10	19–25
J_1	10–17	25–30
J_3	17–23	30–34

Minimum makespan:

$$34$$

(c) The original order J_1, J_2, J_3, J_4, J_5 gives makespan 39.
So Johnson's Rule improves the schedule by

$$39 - 34 = 5$$

time units.

(d) Solving the job shop problem gives an optimal schedule with makespan

$$19$$

One optimal schedule is:

Job	Task	Machine	Start	End
A	1	M_1	0	4
A	2	M_3	5	10
A	3	M_2	13	16
B	1	M_2	0	3
B	2	M_1	6	10
B	3	M_3	10	16
C	1	M_3	0	5
C	2	M_2	5	7
C	3	M_1	10	14
D	1	M_1	4	6
D	2	M_2	7	13
D	3	M_3	16	19

This can be obtained using a job shop formulation or an OR-Tools interval-variable model with precedence constraints within each job and `NoOverlap` constraints on each machine.

(e) Add a due-date constraint for job C:

$$C_C \leq 15$$

where C_C is the completion time of job C.

In the schedule above, job C already finishes at time 14, so this additional constraint is satisfied and one optimal makespan-19 schedule remains feasible. In other instances, such a due-date constraint could force resequencing or even cause infeasibility.

2. Binary Integer Programming

(a) Let $x_i = 1$ if base i is opened and 0 otherwise. Let $y_j = 1$ if district j is covered and 0 otherwise.

$$\max Z = 9000y_1 + 12000y_2 + 14000y_3 + 10000y_4 + 11000y_5 + 8000y_6$$

subject to

$$14x_A + 11x_B + 13x_C + 10x_D + 12x_E + 9x_F \leq 31$$

$$x_A + x_B + x_C + x_D + x_E + x_F \leq 3$$

$$y_1 \leq x_A + x_B$$

$$y_2 \leq x_A + x_C + x_E$$

$$y_3 \leq x_B + x_C + x_F$$

$$y_4 \leq x_A + x_D + x_F$$

$$y_5 \leq x_C + x_D + x_E$$

$$y_6 \leq x_E + x_F$$

$$x_i, y_j \in \{0, 1\}$$

(b) The optimal solution is

$$x_A = x_C = 1, \qquad x_B = x_D = x_E = x_F = 0$$

Total cost:

$$14 + 13 = 27$$

Covered districts:

$$1, 2, 3, 4, 5$$

Total covered population:

$$9000 + 12000 + 14000 + 10000 + 11000 = 56000$$

(c) Yes. In the optimal solution above, district 6 remains uncovered. This is acceptable in a maximum-coverage model because the objective is to maximise covered population subject to tight resources. In a set covering model, complete coverage would be mandatory.

(d) Revised model changes:

District 6 must be covered:

$$y_6 = 1$$

or equivalently

$$x_E + x_F \geq 1$$

Bases B and F cannot both be chosen:

$$x_B + x_F \leq 1$$

Under these revised conditions, one optimal solution is

$$x_E = x_F = 1, \qquad x_A = x_B = x_C = x_D = 0$$

with covered districts $2, 3, 4, 5, 6$ and total covered population

$$12000 + 14000 + 10000 + 11000 + 8000 = 55000$$

There are multiple optimal solutions with the same covered population; for example, $x_C = x_F = 1$ is also optimal.

3. Simulation

(a) Main components:

Entities: students.

Resource: advisers.

Events: arrival, service start, service completion.

Stochastic inputs: interarrival and service times.

Measures: number served, average wait, maximum wait, percentage waiting less than 8 minutes.

(b) One run with seed 24 gave:

Measure	Result
Students served	41
Average wait	1.17 minutes
Maximum wait	10.32 minutes
% waiting less than 8 minutes	95.12%

(c) Peak-period results:

Advisers	Served	Avg wait	Max wait	% below 8 min
2	70	1.88	12.99	95.71%
3	73	0.32	3.22	100%
4	70	0.08	2.23	100%

From this run, the minimum staffing level meeting the target is 3 advisers.

(d) A `PriorityResource` can be used in place of a standard `Resource`. Each arriving student would be labelled priority or standard, for example with probability 0.10 of being priority, and then would request service with a smaller priority number if they are priority users.

In SimPy terms, the main change is from a standard request such as

```
with desk.request() as req:
```

to a priority request such as

```
with desk.request(priority=0) as req:
```

for priority users and a larger number such as `priority=1` for standard users.

This would usually reduce priority-user waits and increase the waits experienced by standard users.

(e) Replications are needed because a single run may be unusually good or bad. If the model represented a continuing system rather than a finite session, a warm-up period might also be appropriate.

4. Forecasting

(a) Using periods 1–12 only, centred 4-quarter moving averages are available for periods 3–10:

Period	Demand	CMA
3	470	491.250
4	560	499.500
5	450	507.750
6	536	516.500
7	500	525.875
8	600	535.125
9	485	544.375
10	575	554.375

A linear trend fitted to these centred moving averages is

$$\hat{T}_t = 463.21 + 9.02t$$

where $t = 1$ corresponds to 2019 Q1.

(b) Seasonal indices:

Additive:

$$Q1 = -58.72, \quad Q2 = 19.91, \quad Q3 = -23.72, \quad Q4 = 62.53$$

Multiplicative:

$$Q1 = 0.8884, \quad Q2 = 1.0372, \quad Q3 = 0.9535, \quad Q4 = 1.1209$$

(c) Validation forecasts for 2022:

Quarter	Actual	Additive Forecast	Multiplicative Forecast
2022 Q1	522	521.76	515.68
2022 Q2	618	609.41	611.44
2022 Q3	572	574.80	570.71
2022 Q4	694	670.07	680.99

Using MAD:

$$\text{MAD}_{\text{add}} \approx 8.89, \qquad \text{MAD}_{\text{mult}} \approx 6.80$$

where absolute error is calculated as $|\text{Actual} - \text{Forecast}|$.

For the multiplicative model:

$$|522-515.68| = 6.32, \quad |618-611.44| = 6.56, \quad |572-570.71| = 1.29, \quad |694-680.99| = 13.01$$

so

$$\text{MAD}_{\text{mult}} = \frac{6.32 + 6.56 + 1.29 + 13.01}{4} \approx 6.80$$

For the additive model:

$$|522-521.76| = 0.24, \quad |618-609.41| = 8.59, \quad |572-574.80| = 2.80, \quad |694-670.07| = 23.93$$

so

$$\text{MAD}_{\text{add}} = \frac{0.24 + 8.59 + 2.80 + 23.93}{4} \approx 8.89$$

The multiplicative model is better.

(d) Forecasts for 2023 using the multiplicative model:

Quarter	Forecast
2023 Q1	547.74
2023 Q2	648.86
2023 Q3	605.11
2023 Q4	721.43

5. Linear Programming

Let C be casings and I be inserts.

(a) A suitable model is

$$\max Z = 40C + 34I$$

subject to

$$4C + 2I \leq 100$$

$$C + 3I \leq 90$$

$$I \leq 24$$

$$C, I \geq 0$$

(b) The optimal solution is

$$C = 13, \qquad I = 24$$

Maximum contribution:

$$Z = 40(13) + 34(24) = 1336$$

(c) Binding constraints: moulding and insert demand cap.

Slack values:

$$4(13) + 2(24) = 100 \Rightarrow 0 \text{ moulding slack}$$

$$13 + 3(24) = 85 \Rightarrow 5 \text{ assembly hours unused}$$

$$24 - 24 = 0$$

(d) The new condition is

$$I \geq 0.25(C + I)$$

which simplifies to

$$C - 3I \leq 0$$

At the original optimum,

$$13 - 3(24) = -59 \leq 0$$

so the original solution remains feasible and optimal.

(e) With 12 extra assembly hours, the assembly constraint becomes less restrictive, but the current solution already has 5 hours of assembly slack and is limited by moulding plus the insert cap. The optimal plan therefore does not change.

Extra contribution gained:

$$0$$

Maximum cost of buying all 12 hours:

$$12 \times 3 = 36$$

So the extra hours should not be bought.

6. Transportation

(a) Let x_{ij} be the quantity shipped from warehouse i to store j.

A suitable model is

$$\min Z = \sum_i \sum_j c_{ij} x_{ij}$$

subject to the supply constraints for each warehouse i:

$$\sum_j x_{ij} = s_i$$

demand constraints for each store j:

$$\sum_i x_{ij} = d_j$$

and non-negativity:

$$x_{ij} \geq 0$$

(b) One optimal plan is:

Route	Quantity
$W_1 \to S_1$	20
$W_1 \to S_4$	60
$W_2 \to S_2$	60
$W_2 \to S_3$	40
$W_3 \to S_1$	50
$W_3 \to S_3$	40

Total transportation cost:

$$1560$$

(c) With $W_2 \not\to S_2$ and at least 20 units from W_1 to S_4, one optimal revised plan is:

Route	Quantity
$W_1 \to S_2$	60
$W_1 \to S_4$	20
$W_2 \to S_3$	60
$W_2 \to S_4$	40
$W_3 \to S_1$	70
$W_3 \to S_3$	20

Revised cost:

$$1680$$

Increase in cost:

$$1680 - 1560 = 120$$

(d) Example extension: add vehicle capacities or delivery time windows so the model reflects route feasibility as well as shipping cost.